Analyzing Socioeconomic Factors Impacting Business Closure Rates in San Francisco During and Post-COVID-19

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Introduction:

The COVID-19 pandemic had a significant economic impact on businesses around the world, causing many to shut down permanently. One location which was especially impacted was the city of San Francisco. During the COVID-19 pandemic and in the years since, San Francisco has experienced economic difficulties, which have contributed to high rates of business closure. However, certain areas of the city experienced much higher rates of closure than others. The question I attempted to answer in this analysis is whether there is a notable connection between the socioeconomic characteristics of an area (Median income, Property value, racial makeup, total population) and the rate of business closure in that area.

A similar published paper, written by Yasuyuki Motoyama, examined whether COVID-19 had a disproportionate impact on “Poor and Minority neighborhoods” [1]. This study compared business closure rates by census tract and neighborhood to the rates of poverty, racial makeup, and rates of college education in the same areas using Regression Analysis. Contrary to the results of my analysis, this analysis found that there was no significant correlation between these variables. This difference may be caused by the fact that this paper spent more time ensuring that there were no biases in the data (For example, the paper criticizes the use of census tracts as a geographical unit due to their small size), or by the fact that this paper focused on the COVID-19 pandemic as a whole, rather than just the San Francisco area, so it is possible that the results could differ for legitimate reasons.

In this paper, I will first show visualizations of all significant variables by census tract. Next, I will show the results of both correlation coefficient and clustering analysis of this data. Lastly, I will summarize the results of the analysis, and discuss the possible implications of these results.

Data:

I gathered data on median household income, median property value, total population, and white, black, asian, and hispanic/latino populations from the American Community Survey [2] and the Decennial Census [3]. I also used a table from datasf [4] containing the locations, opening and closing dates of all businesses in San Francisco, which I used to obtain the statistic of what percentage of businesses closed between 2020 and 2024 in each San Francisco census tract.

Spatial Visualization of Main Variables:

*Higher values of all variables are represented by brighter colors (yellow), and lower values by darker colors (purple)

** Additional visualizations can be found in the appendix

Figure 1. Percent of SF businesses closed between 2020 and 2024 by tract (Percent_Closed)

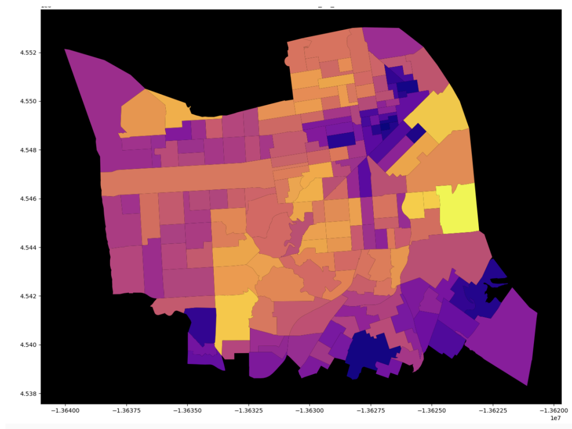

Figure 2. Median Household income by tract (median_hh_income)

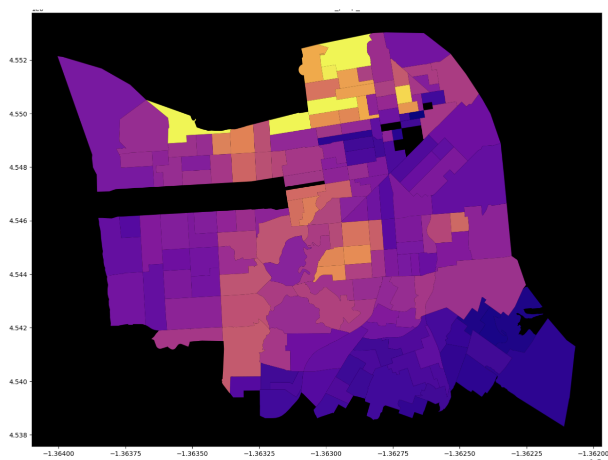

Figure 3. Median property value by tract (median_prop_value)

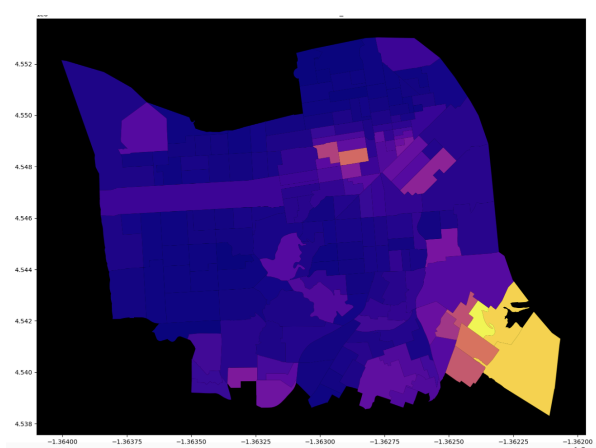

Figure 4. Black population percentage by tract (percent_black)

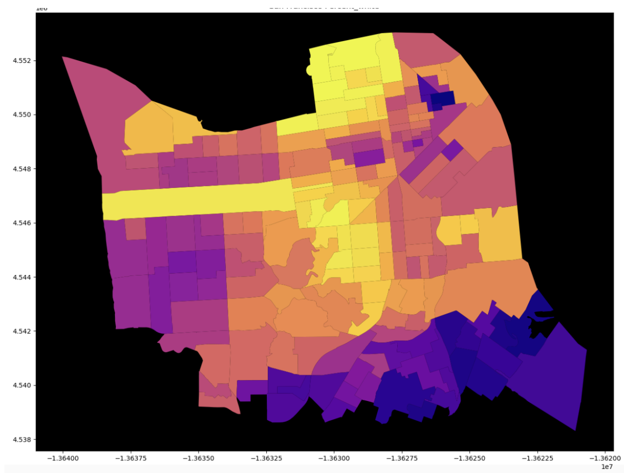

Figure 5. White population percentage by tract (percent_white)

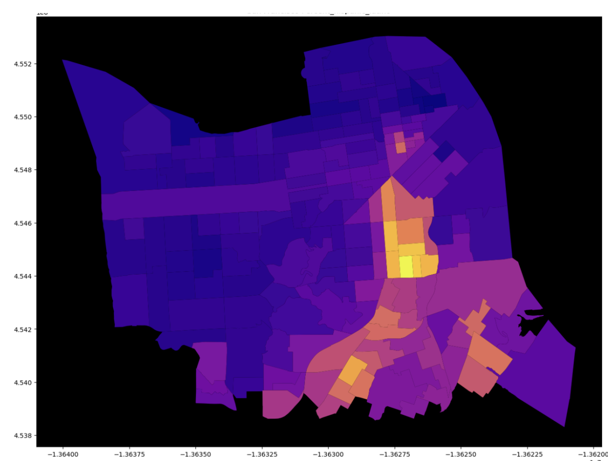

Figure 6. Hispanic/Latino population percentage by tract (percent_hispanic_latino)

Correlation Coefficients:

Another method of analysis I used on the data was the correlation coefficient between each pair of variables, especially the correlation between percent_closed and each other variable. Unfortunately, this analysis was mostly inconclusive, with little correlation between most variables. However, there was some correlation between percent_closed and racial demographics, with slight positive correlation between percent_closed and both percent_black and percent_hispanic_latino, and a slight negative correlation between percent_closed and percent_white.

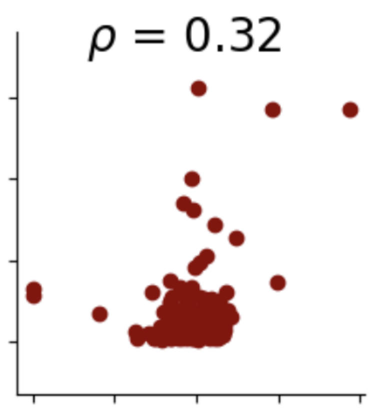

Figure 7. Percent_closed (x) vs. percent_black (y)

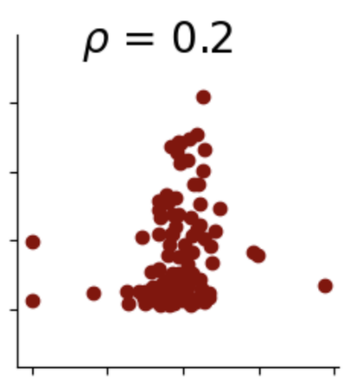

Figure 8. Percent_closed (x) vs. percent_hispanic_latino (y)

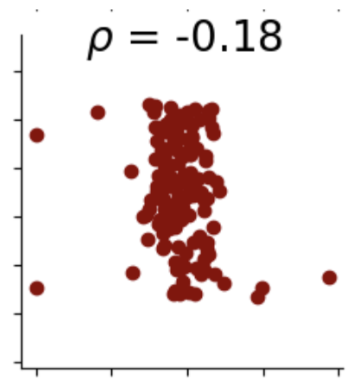

Figure 9. Percent_closed (x) vs percent_white (y)

K-means Clustering:

My final method of analysis on this data is K-means clustering. My goal for the clustering was to summarize the general characteristics of regions of San Francisco with different rates of business closure. Based on the inertia and variable distributions of each possible number of clusters, I determined 3 clusters was the most effective number for this data.

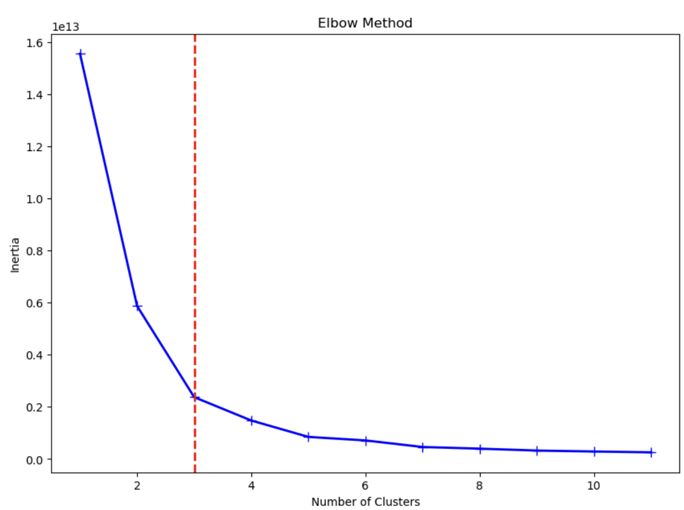

Figure 10. The elbow method applied to my data

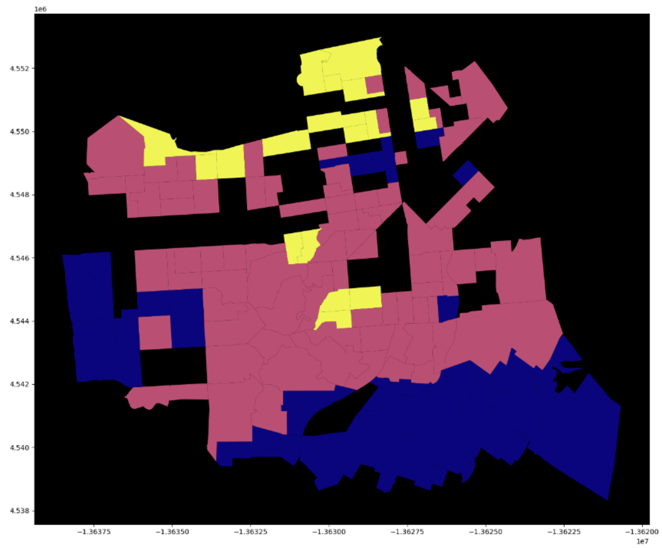

Figure 11. Cluster by tract (Blue - Cluster 0, Pink - Cluster 1, Yellow - Cluster 2)

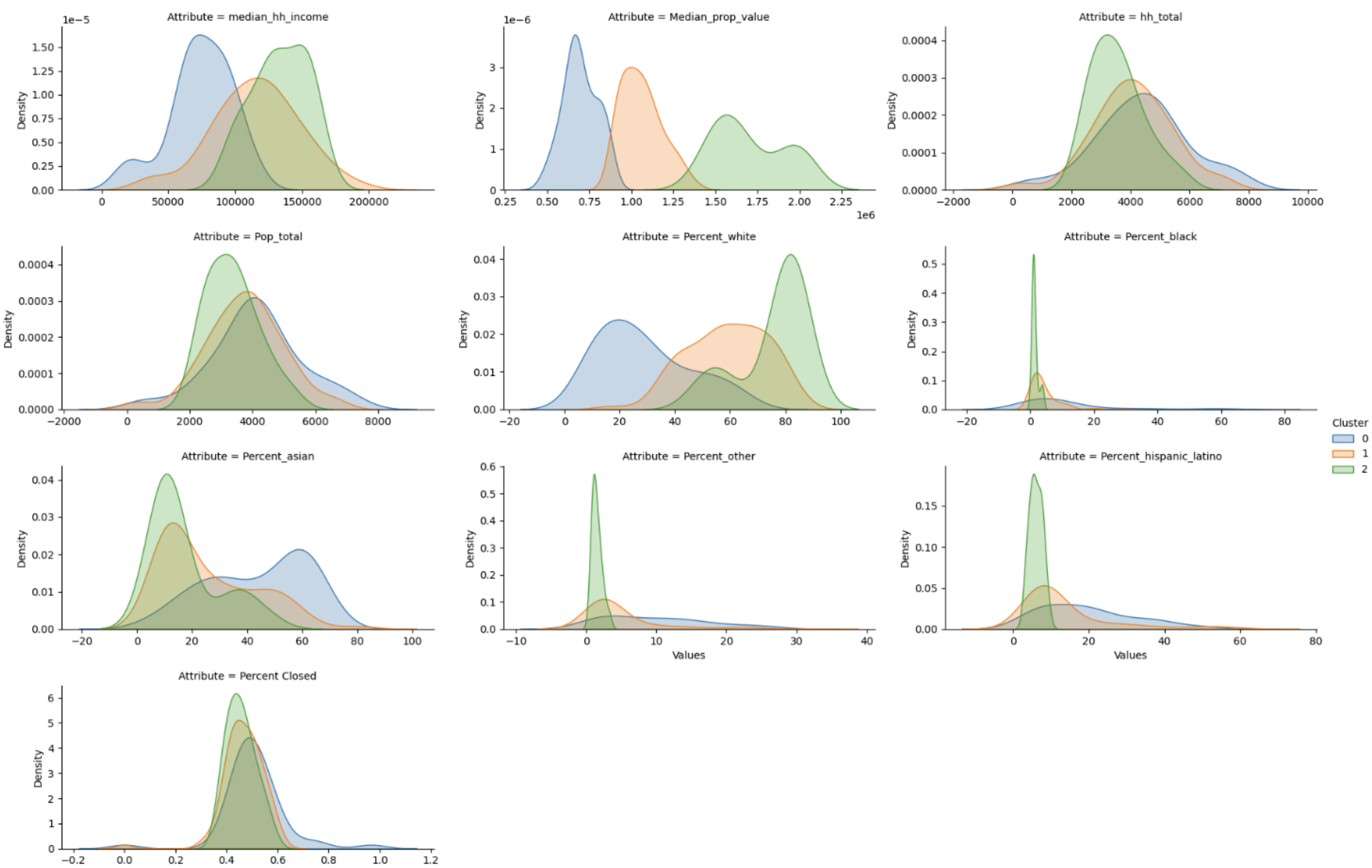

Figure 12. Plots of each variable by cluster

Based on this data, I determined the general characteristics of each cluster.

**Average rate of business closure across all clusters is 47.63%*

Cluster 0 is mainly lower income and property value, with a lower white population, with higher percentages of African Americans, Hispanic/Latinos, and Asians. It also has a higher overall population, and has the highest rate of business closure, with a mean of 50.667%.

Cluster 1 is mostly middle income and property value, with a population that, while mostly white, is significantly more diverse than cluster 2. It has a slightly lower population than Cluster 0, and a lower rate of business closure, with a mean of 46.374%.

Cluster 2 contains higher income levels and significantly higher property values, as well as a significantly lower population. Its population is almost entirely white, with a non white population that is mostly Asian. This cluster has the lowest rate of business closure, with a mean of 45.616%

Conclusion:

The Correlation Coefficient and k-means Clustering analysis suggest that there is a small but notable difference in rates of business closure between areas of San Francisco with different socioeconomic characteristics. Proportionally larger non-white populations, lower incomes/property value, and higher population are associated with slightly higher rates of business closure, while proportionally higher white populations, higher income/property value, and lower population are associated with lower rates. The results of this analysis could be improved by improving the quality of the data, such as by determining a more specific timeframe in which to examine business closures, or by using other geographic metrics. These results could also be compared to other economic crises throughout the world and/or in the past, to gain a better understanding of what causes these results, and what could be done in the future to remove this inequity.

Appendix: Other figures

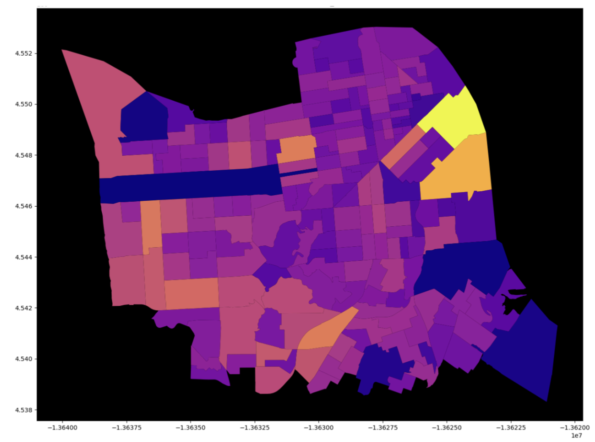

Figure 13. Total households by tract (hh_total)

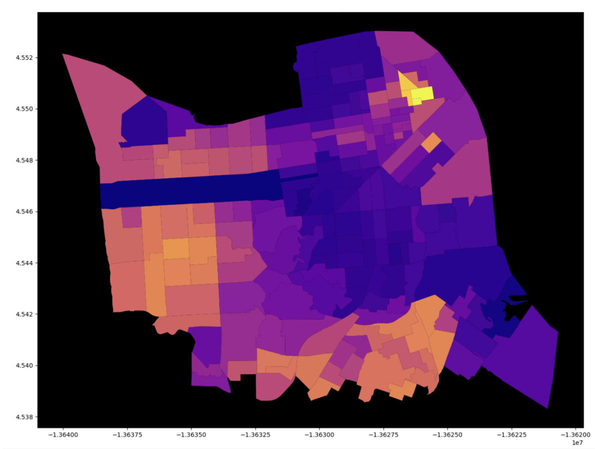

Figure 14. Asian population percentage by tract (percent_asian)

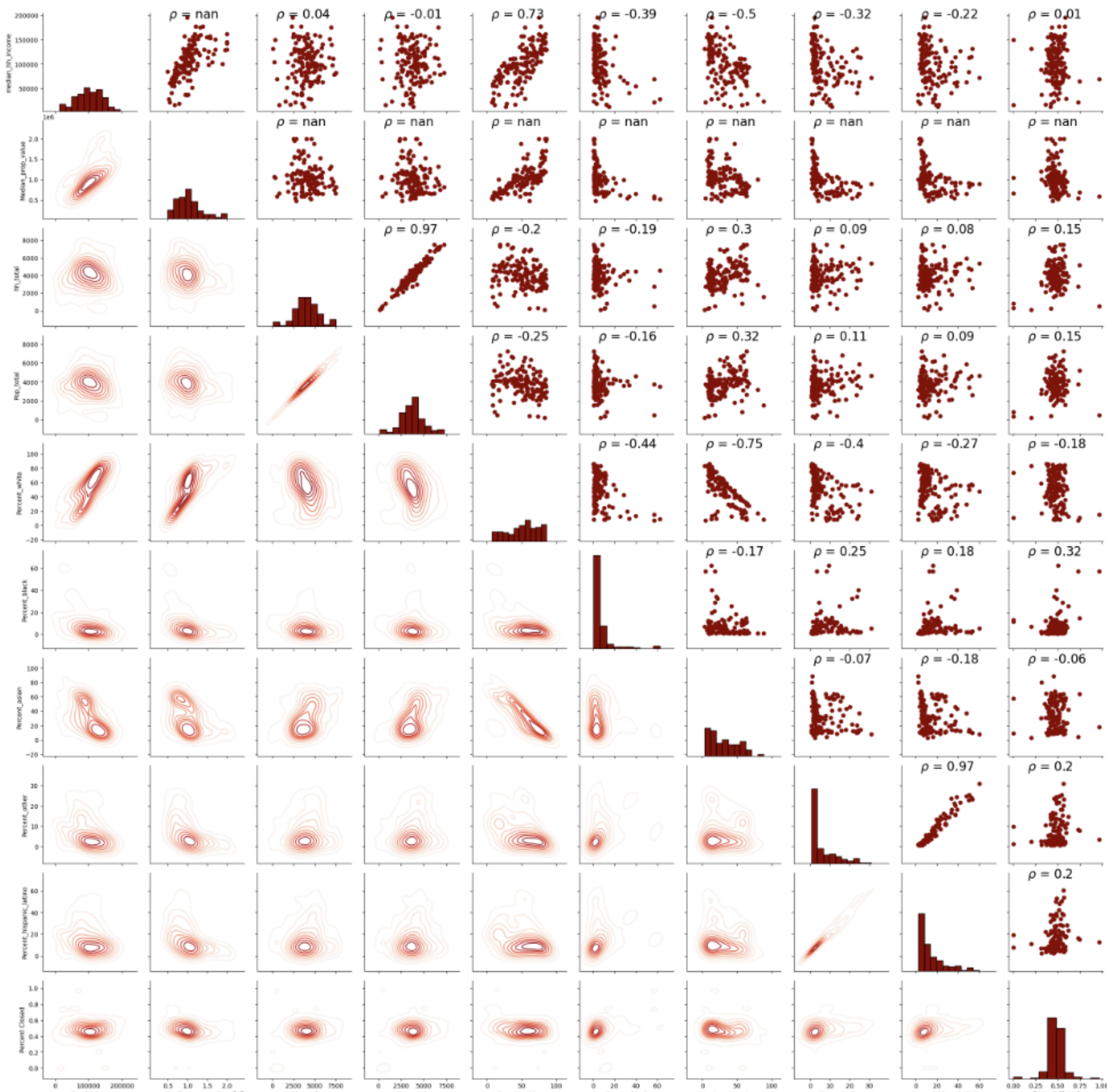

Figure 15. Correlation scatter plots comparing every pair of variables

References:

[1] Motoyama YY. Is COVID-19 Causing More Business Closures in Poor and Minority Neighborhoods? Econ Dev Q. 2022 May;36(2):124-133. doi: 10.1177/08912424221086927.

PMID: 37519428; PMCID: PMC8922045,
https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8922045/

[2] “Census Data API: /Data/2017/ACS/acs5/Variables.”Developers, 28 Nov. 2011,
https://api.census.gov/data/2017/acs/acs5/variables.html

[3] “Census Data API: /data/2010/dec/sf1/variables” Developers,
https://api.census.gov/data/2010/dec/sf1/variables.html

[4] “San Francisco Open Data.” *DataSF*, https://datasf.org/opendata/. Accessed 28 Apr. 2024.